

Index of a closed Curve.

Let $\gamma(t) = a + e^{2\pi i n t}$ ($0 \leq t \leq 1$). Then,

$$\int_{\gamma} \frac{1}{z-a} dz = \int_0^1 e^{-2\pi i n t} \cdot 2\pi i n \cdot e^{2\pi i n t} dt = 2\pi n \cdot i.$$

In general,

Lemma. Let $\gamma: [0,1] \rightarrow \mathbb{C}$ be a closed rectifiable curve and $a \notin \text{Im } \gamma$, then

$$\frac{1}{2\pi i} \cdot \int_{\gamma} \frac{1}{z-a} dz$$

is an 'integer'.

Proof. Assume that γ is a smooth map. Define $g: [0,1] \rightarrow \mathbb{C}$ as

$$g(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s)-a} ds$$

Then, $g(0) = 0$ and $g(1) = \int_0^1 \frac{\gamma'(s)}{\gamma(s)-a} ds = \int_{\gamma} \frac{1}{z-a} dz$

And $g(t) = \frac{\gamma(t)}{\gamma(t)-a}$ for $0 \leq t \leq 1$.

But, $\frac{d}{dt} e^{-g(t)} \cdot (\gamma(t)-a) = e^{-g(t)} \cdot \gamma'(t) - g'(t) \cdot e^{-g(t)} \cdot (\gamma(t)-a) = 0$

So, $e^{-g(t)}(\gamma(t)-a)$ is the constant,

$$e^{-g(0)}(\gamma(0)-a) = \gamma(0)-a = e^{-g(1)}(\gamma(1)-a)$$

Since γ is closed curve, so $\gamma(1) = \gamma(0)$,

$$e^{-g(1)} = 1 \quad \text{or} \quad g(1) = 2k\pi \cdot i \quad \text{for some } k \in \mathbb{Z}.$$

Def. If γ is a closed rectifiable curve in $\mathbb{C}$ and $a \notin \text{Im } \gamma$,

$$n(\gamma; a) \stackrel{\text{def}}{=} \frac{1}{2\pi i} \cdot \int_{\gamma} \frac{1}{z-a} dz$$

is called the index of γ with respect to a .